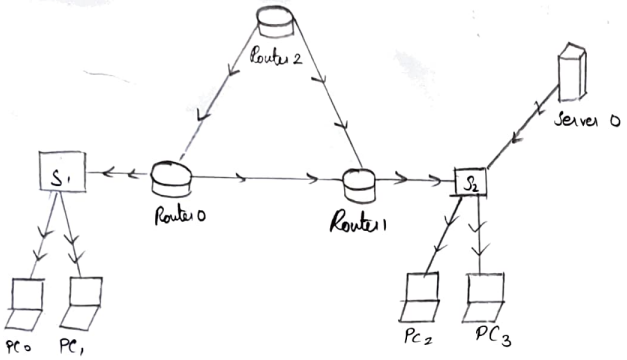


ROUTER	Available networks on local interfaces	Networks available on other routers' interfaces
Router 0	10.0.0.0/8, 20.0.0.0/8, 40.0.0.0/8	30.0.0.0/8, 50.0.0.0/8
Router 1	20.0.0.0/8, 30.0.0.0/8, 50.0.0.0/8	10.0.0.0/8, 40.0.0.0/8
Router 2	40.0.0.0/8, 50.0.0.0/8	10.0.0.0/8, 20.0.0.0/8, 30.0.0.0/8.

EXP 11- ROUTING AT NETWORK LAYER:

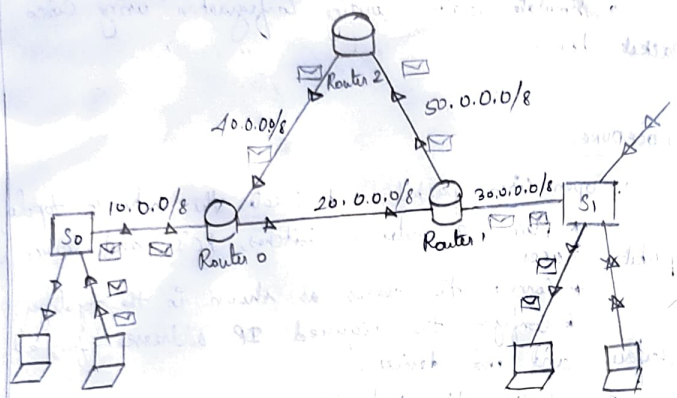
(a)

AIM:

To simulate static routing configuration using Cisco packet tracer.

PROCEDURE:

1. Open Cisco packet and create the network topology.
   * Place 3 routers, switches, PC's and server in packet tracer.
   * Connect the devices as shown in the topology.
   * Configure the required IP addresses for all routers interfaces and end devices.
2. Identify the Networks.
   * The networks used are: 10.0.0.0/8, 20.0.0.0/8, 40.0.0.0/8 and 50.0.0.0/8.
   * Identify the remote networks that are not directly connected to each router.
3. Configure Static Routes.
   * Open each router → CLI
   * Enter configuration mode:
     ↳ enable
     ↳ configure terminal.
4. Configure Main and Backup Routes.
   * Configure a primary route with a lower administrative Distance (AD).
   * Configure another route with a higher AD as the backup.
   * If the main route fails, the router automatically uses the backup route.
5. Verify static Routing.
   * Check configured static routes using.
   * Test Communication using ping and tracert.
   * Disconnect the main route and verify that packets travel through the backup route.



6. Delete a static Route if required.

→ no ip route <destination - network> <subnet - mask> <next - hop>.

RESULT:

Thus, static routing with main and backup routes was successfully configured using Cisco Packet Tracer. Connectivity between different networks was verified using Ping and tracert and the backup route was successfully used when the main route failed.